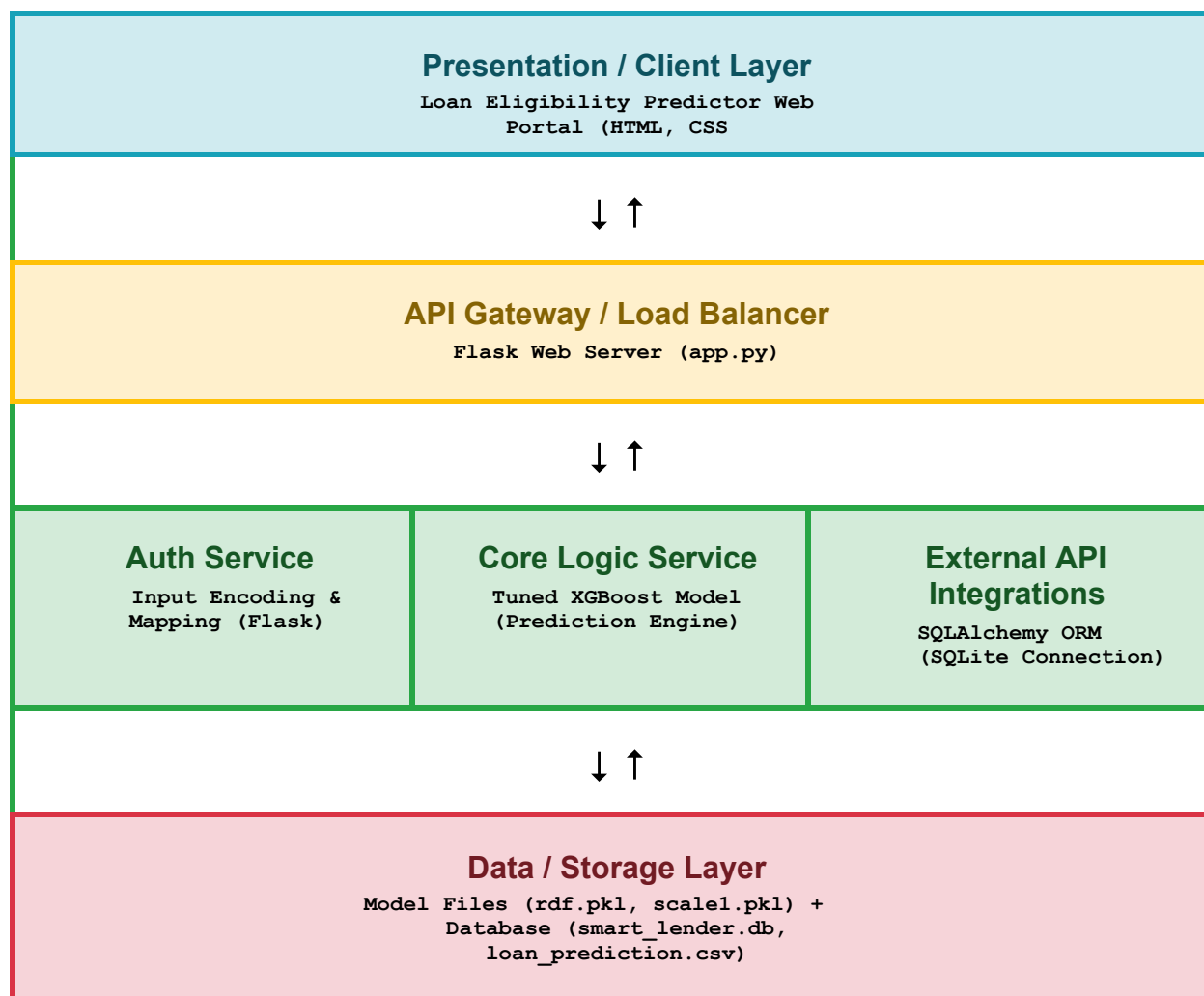


Solution Architecture

Date	05 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where loan officers input candidate details (income, credit score, background) and view eligibility status results.	HTML5, CSS3 (Glassmorphism layout), JavaScript
API Gateway	Flask receives client HTTP requests, validates and maps categorical inputs to integers, and routes requests to the model and database.	Python, Flask
Core Logic Service	Standardizes numerical fields using the saved scale1.pkl scaler, processes prediction parameters using the tuned XGBoost model (rdf.pkl), and calculates the loan approval probability.	Scikit-learn, XGBoost, Pickle
Data Layer	Stores the serialized model binaries (rdf.pkl, scale1.pkl), the local transaction database (smart_lender.db), and the original training dataset.	Pickle (.pkl), SQLite (.db), SQLAlchemy, Pandas (.csv)